

Strategic Patent Listing and Generic Challenge in China’s Drug Patent-Linkage Register

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Abstract

China’s 2021 patent-linkage reform created the world’s first multi-class register of drug–patent linkages, and we observe it from birth—a complete, dated census of both sides of an Orange-Book-style institution at day one. We provide an anatomy. Originators list selectively (listed chemical patents carry a $3.7\times$ family-citation premium; biologics $11\times$), yet the register’s backbone is thin: 45% of listed chemical products have no live compound patent, and secondary patents outlive compounds by a median five years. Challenges are 99% chemical, arrive ten months after listing, fall twice as often on compound-free products, and target foreign originators; their post-reform rise tracks a gradual ramp, not a dated response to the first judicial decision. Listed patents have begun to die by invalidation—84% of them on drugs already under challenge—yet the register does not notice: three dozen prematurely dead patents remain on display with stated expiries reaching 2038. Equally diagnostic is what the register cannot yet identify: no product-specific defensive-listing response to challenges, patent-wall deterrence changes sign across challenge measures, and statutory class machinery is

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not separable from the adoption ramp. Hatch–Waxman’s *static* strategic grammar—selective listing and quality-sorted challenge—travels within five years under a far weaker reward; its dynamic responses, as the nulls above show, do not.

Keywords: patent linkage; generic entry; secondary patents; evergreening; pharmaceutical regulation; China.

JEL: K11; L65; O34; I18.

1 Introduction

Forty years after Hatch–Waxman ([Hemphill and Sampat, 2025](#)), nearly everything economists know about patent-linkage institutions—which patents originators list, which listings generics attack, what the resulting bargaining does to entry—comes from a single register observed decades into its operation, with the survivorship, left-truncation, and accumulated case law that maturity implies. In July 2021 China switched on a major new linkage register, and the first to pool chemical drugs, biologics, and traditional Chinese medicine (TCM) under one statute. We observe it from birth: every listing event, every update, and every generic challenge declaration, timestamped, from day one through its first five years. This paper is an anatomy of that institution—a set of sharply bounded descriptive and quasi-experimental facts about how a linkage register populates, what it protects, and whom its challenges target, in the tradition of [Hemphill and Sampat \(2011\)](#) and [Hemphill and Sampat \(2012\)](#) for the Orange Book.

A single statute, we will show, houses two economically distinct institutions. For chemical small molecules the register is *litigation infrastructure*—selective listing, secondary-patent walls, and a quality-sorted challenge stream, the object of this paper. For traditional Chinese medicine (TCM), a class that faces no functioning challenge apparatus and is challenged at a fraction of the chemical rate, voluntary listing cannot be litigation readiness; the register there operates as a *certification platform*, the subject of a companion paper ([\[Author](#)

[anonymized for peer review](#)], 2026). Watching one register divide into two under formally identical procedure is a contrast the mature, combat-only Orange Book cannot supply, and it is the frame within which the facts and nulls below are read.

Five facts organize the anatomy. *First, listing is quality-selective, and selective on a global margin.* Within narrow technology–vintage–applicant cells of a 1.56-million-patent pharmaceutical universe, listed chemical patents carry a $3.7\times$ family-citation premium over unlisted peers and biologics $11.2\times$, while TCM shows no family premium at all—yet all three classes select on domestic citations. The register recruits globally defended portfolios exactly where the statutory machinery binds. *Second, the register’s protective backbone is thin.* Forty-five percent of listed chemical products carry no live compound patent, and where both patent types are listed the secondary patent outlives the compound by a median five years—an evergreening structure assembled at birth, not accreted over decades. *Third, challenges chase weakness and foreignness.* Generic challenges are 99% chemical, arrive a median ten months after listing, fall twice as often on products with secondary-only protection, and target foreign originators’ products at rates 23–42 points higher—while every challenger in the data is domestic. *Fourth, challenge filing ramps up after the reform but not at a datable moment:* chemical challenge flows are roughly 70% higher after the first linkage decision, but the rise is gradual—a placebo break a quarter earlier is equally significant and the level shift does not survive a flexible trend—so we read it as post-reform ramp-up, not a discrete enforceability-news jump. *Fifth, listed patents have begun to die by invalidation—and the register does not notice.* Forty-four listed patents have been fully invalidated, all but two post-reform; 84% of them protected drugs already under generic challenge (median lag twenty-one months from first challenge to invalidation, against a 29% baseline challenge incidence), whereas patents that lapse for non-payment of fees are almost never challenged first. The enforcement margin therefore works—through the patent office, not the courts. The bookkeeping does not: thirty-six prematurely dead patents remain listed with stated expiries reaching 2038.

A note on language: throughout, “select,” “chase,” and “target” name conditional associations in observational data, not demonstrated intent; we adopt them for economy, and each causal margin below is either identified by a pre-committed test or explicitly declined.

The anatomy is disciplined by what we can show is *not* there, and we report these nulls as findings. Originators do not update listings in response to their own challenges—the apparent post-challenge spike is fully reproduced when challenge dates are randomly reassigned within calendar quarters; whether denser patent walls deter challenges cannot be answered in this register—the patent-count gradient flips sign across the two available challenge measures at every level of aggregation, so we decline the Ellison–Ellison question (Ellison and Ellison, 2011) rather than answer it badly; and the class-specific statutory machinery, the most tempting source of causal identification, fails its falsification—biologic listing flows respond to “chemical-only” judicial milestones as strongly as chemical flows—so cross-class contrasts carry descriptive weight only.

These facts speak to three literatures. The strategic-listing and challenge-targeting results give out-of-sample discipline to the U.S. evidence that challenges concentrate on weak secondary patents covering valuable products (Hemphill and Sampat, 2012) and to the policy concern with improper listing (Federal Trade Commission, 2023): China’s register reproduces the secondary-patent attraction margin within five years of existence, under a different legal system, with a narrower set of listable categories. The selection results contribute to the patent-quality measurement literature (Trajtenberg, 1990; Hall et al., 2005; Sampat and Williams, 2019) a clean demonstration that family- and domestic-citation measures identify different selection margins when classes differ in international-filing propensity. And the institutional results inform the comparative welfare question of what linkage does to entry (Branstetter et al., 2016): we characterize the conditions—thin compound backbones, reactive late listing, an accelerating post-reform challenge stream—under which China’s much weaker exclusivity reward (twelve months, nine-month stay) is nonetheless generating Paragraph-IV-like behavior. The margin where that weaker reward most plausibly bites is

the settlement bargain—the pay-for-delay problem that dominates the U.S. welfare debate (Hemphill, 2006; Supreme Court of the United States, 2013)—since a twelve-month exclusivity and a nine-month stay set the stakes of any challenge-and-settle negotiation; that bargain is not yet observable in this register.

This paper is closest to a small descriptive literature on China’s register: Yao (2024), Han (2022), and Zhang et al. (2024) document the mechanism’s early operation, and Li and Teng (2025) profile the register’s patent-portfolio composition; methodologically it is adjacent to work on the timing of Orange Book listings in the United States (Latarnser et al., 2026). We differ on four margins. We build a from-birth, *dated* census of both sides of the register; we estimate quality selection against a 1.56-million-patent pharmaceutical denominator; we document register integrity—invalidations and premature-death contradictions the register never records—as a first-order fact; and we read the whole through an industrial-organization lens disciplined by pre-committed falsifications rather than as a policy description.

The rest of the paper proceeds as the anatomy requires: §2 sets out the institution, §3 a minimal framework that generates the facts and reads the nulls, §4 the three-layer data architecture, §5 the supply side (selection and composition), §6 listing dynamics, §7 the demand side (timing, hazards, and targeting), §8 the nulls, and §9 concludes.

Table 1: The five facts at a glance. Every figure is established and sourced in the section named; the paper’s discipline is the set of nulls (§8) that bound them.

Fact	Where
1 <i>Quality-selective listing.</i> Listed chemical patents carry a $3.7\times$ family-citation premium, biologics $11\times$ (coverage-matched +247%/ +990%); TCM shows no family premium.	§5.1
2 <i>A thin protective backbone.</i> 45% of listed chemical products have no live compound patent; where both are listed the secondary patent outlives the compound by a median 5 years.	§5.2
3 <i>Challenges chase weakness and foreignness.</i> 99% chemical; secondary-only products +14–17pp more likely challenged; foreign originators’ products +23–42pp.	§7.4
4 <i>A post-reform ramp, not a dated jump.</i> Chemical challenge flows are $\sim 70\%$ higher after the first linkage decision, but the break is not datable (fails a placebo-in-time test).	§7
5 <i>Register integrity.</i> 44 listed patents fully invalidated (84% challenged first); 36 prematurely dead patents remain listed with stated expiries reaching 2038.	§5.2

2 Institutional Background

2.1 The 2021 linkage system

China’s path to a functioning linkage system unfolded in three stages: the 2017 “Opinion 42” directive to explore a patent-linkage system (Liang, 2018); the October 2020 Fourth Amendment to the Patent Law, whose Article 76 created a statutory cause of action for disputes arising during drug marketing approval; and the operative stage on 4 July 2021, when the joint NMPA–CNIPA *Implementing Measures for the Early Resolution Mechanism of Pharmaceutical Patent Disputes (Trial)* established the CDE patent-information registration platform and the four-category declaration scheme (Han, 2022; Yao, 2024).

Four features differentiate China from the Hatch–Waxman benchmark. *First*, the CDE

platform is a public, searchable, Orange-Book-like register maintained for chemical drugs, biologics, *and* TCM—the only multi-class register in the world. *Second*, generic and biosimilar applicants must file one of four declaration types; only Type IV (non-infringement or invalidity) triggers the stay. *Third*, the stay is nine months—far shorter than the U.S. 30 months—running in parallel with dual-track administrative (CNIPA) and judicial (Beijing IP Court, SPC IP Tribunal on appeal) proceedings (Han, 2022). *Fourth*, the first successful challenger earns a twelve-month market exclusivity whose qualification criteria remain ambiguous in practice (Zhang et al., 2024). Listing is self-certified in both systems—the margin on which register-integrity concerns operate (Federal Trade Commission, 2023). Liu (2016) documents that a formally more protective 2002 linkage regulation failed operationally; 2021 is China’s second attempt, which motivates attention to implementation detail over statutory design.

2.2 The chemical-only statutory machinery

The stay-and-exclusivity machinery is, in its operative detail, built around chemical small-molecule products and their ANDA-style challengers; the biosimilar and TCM pathways carry no equivalently developed challenge-and-stay apparatus (Han, 2022; Yao, 2024). TCM has no Hatch–Waxman analogue anywhere in the world (Son et al., 2018), yet TCM patents sit on the same register under the same declaration scheme—a within-institution comparison class. We are explicit about how much weight this asymmetry can bear: it organizes *levels* (challenges are 98.95% chemical; bio/TCM challenge hazards are near zero), but it does not support causal difference-in-difference claims, because biologic listing flows respond to “chemical” milestone news as strongly as chemical flows do (§8). We therefore use the asymmetry descriptively throughout, and the selection result of §5.1—the largest quality premium is biologic, not chemical—already points away from a pure stay-defense mechanism toward portfolio signaling and global-breadth selection. TCM enters this paper only as a scale reference; its own listing behavior—voluntary registration under no credible challenge

threat—is analyzed in the companion paper ([Author anonymized for peer review], 2026), which keeps the cross-class comparison here to levels and out of the causal claims.

2.3 Milestones

Table 2 collects the dates that matter for the analysis. We emphasize one design decision: the platform launch (July 2021) is the *sample start*, not an analyzable event—pre-launch register flows are mechanically zero—so the informative milestones are the first judicial decision under the new mechanism and the patent-term-extension regime.

Table 2: Institutional milestones (primary-sourced in the replication package). The launch composite defines the sample start; Chugai and PTE are the analyzable events (§7).

2020-10-17	Patent Law 4th Amendment adopted (Art. 76 cause of action)
2021-06-01	Amended law effective
2021-07-04/05	Implementing Measures; CDE platform live; SPC interpretation; CNIPA administrative measures (composite; = sample start)
2022-04-15	<i>Chugai v. Wenzhou Haihe</i> (eldecalcitol), Beijing IP Court first instance — first decision under the linkage mechanism
2022-08-05	SPC final judgment in <i>Chugai</i>
2024-01-20	Patent-term-extension regime effective

3 A Simple Framework

The anatomy is organized by a minimal model of the two choices the register records—which patents an originator lists, and which listed products a generic challenges. We do not estimate it structurally; its role is to state, before the data, why the facts below are the ones an economist should expect, and to make the nulls interpretable as failed predictions rather than as missing results.

The listing choice. An originator holding a granted patent on an approved product lists it when the value of registration exceeds its near-zero cost. That value has two components. A *defense* value—the option to trigger the nine-month stay and contest a generic challenge—is realized only where a challenge apparatus exists (chemical small molecules) and loads on *globally defended* patents, those with wide international families, the breadth on which market-defense calculus operates. A *certification* value—the public, state-backed signal that the patent protects the product, addressed to procurement bodies, formularies, and rivals—requires no challenge threat. The model thus predicts (i) *quality-selective listing*: listed patents are positively selected on quality relative to comparable unlisted patents; and (ii) that the selection loads on the *global-family* margin precisely in the classes where defense value binds (chemical, biologic), and on the *domestic* margin for the class where only certification value operates (TCM). The selection premia and the family-versus-plain divergence of §5.1 are these two predictions.

The challenge decision. A generic challenges a listed product when its expected payoff—the product’s market value times the probability of clearing the listed patents, net of challenge cost—is positive. Two comparative statics follow. First, the clearing probability is higher where protection rests on *secondary* (formulation, use) patents rather than the compound, which are weaker and more often invalidated, so challenges concentrate on secondary-only products (the attraction margin, §7.4). Second, holding weakness fixed, the highest-value molecules in this register are the import-registered products of *foreign* originators, against whom every challenger in the data is a domestic entrant; challenges therefore skew foreign (the targeting margin). Whether that skew is a pure value effect or a distinct origin premium is separable, and §7.4 shows it is the latter: the foreign gradient survives—indeed strengthens under—an external reimbursement-value control, so value-targeting and foreign-targeting are separate margins rather than one dressed as the other.

What the model does not predict. The static model is silent on dynamics, and this is where the register’s infancy bites. It gives no re-optimization margin once a patent is listed at near-zero cost, so it predicts no defensive re-listing in response to a challenge (§6); it does not sign the patent-*count*-challenge relationship, since more patents raise both the value signaled and the cost of clearing (the indeterminate deterrence gradient); and it says nothing about the *timing* of challenge flows, which the data attribute to platform adoption rather than to datable enforceability news (§8). The framework thus generates the paper’s positive facts as predictions and its nulls as silences—each a place the static grammar of listing-and-challenge does not reach.

4 Data

We assemble three layers, joined on normalized patent and approval numbers. For China this construction is the analogue of the NBER Orange Book dataset for the United States (Durvasula et al., 2023)—a research-grade census of a linkage register—extended with a dated event layer and a matched unlisted comparison universe. (i) *The register itself*: a complete scrape of the CDE patent-information platform—7,905 listing rows covering 2,006 distinct patents and 1,821 registered products, with a dated event layer (3,642 registration events, 首次 vs. 更新) that timestamps every listing and update; 948 products carry dated listed patents and constitute the risk set. Each register row carries the platform’s own patent-type category, which we collapse to the standard primary/secondary dichotomy (2,485 primary active-compound or sequence-structure patents vs. 6,278 secondary formulation/use/extract patents across 8,778 rows; IPC cross-validation agreement 88.9%) and the originator-stated expiry date (99.8% coverage). (ii) *The full CNIPA corpus* (57.3M patents, 1985–July 2025), from which a six-prefix pharma-IPC universe of 1.56M patents provides the unlisted comparison set and citation outcomes (Trajtenberg, 1990; Hall et al., 2005). (iii) *The challenge stream*: 17,599 generic-challenge declarations scraped from CDE acceptance records, joined

to products two ways—a strict approval-number join and a name-band (molecule-level) join that bound true incidence from below and above. All product-level results are reported under both joins; the band is the headline outcome and strict the robustness column throughout. From these we build a product master and a product×month hazard panel (41,597 rows) with time-varying portfolio covariates.

Coverage: the central data caveat, up front. The CNIPA merge has a systematic, class-differential non-match that we treat as a first-order identification issue. Of 2,006 distinct CDE patents, 1,233 (61.5%) match the corpus; the 774 misses are overwhelmingly PCT national-phase entries absent from the CNIPA snapshot, and they are concentrated by class—match rates of 50.4% (chemical), 36.8% (biologic), 97.5% (TCM)—so the matched chemical and biologic samples are effectively domestic-route samples stripped of their highest-citation foreign members. Three responses: (i) a normalization audit rules out a key-construction artifact (0 of 774 recovered under identical two-sided normalization; a positive control recovers 94%); (ii) a *coverage-matched* specification restricting both sides to domestic-route patents is a primary specification, not a robustness afterthought; (iii) we measure the missing patents *directly*: a two-source supplement (Google Patents; EPO Open Patent Services) resolves 760 of the 779 unmatched patents (Appendix B). Calibrated to the CNIPA citation scale through a dual-measured pilot (Spearman $\rho = 0.90$), the excluded chemical patents average **4.3 times** the matched-sample family-citation mean and biologics **3.5 times**, with a median INPADOC family spanning 27 jurisdictions—so the omission biases the measured selection premium *toward zero*, and the reported chemical and biologic premia are lower bounds: a statement we previously signed and can now quantify. Register-side analyses (composition, §5.2; timing; hazards; targeting) use the register’s own fields (type, expiry, dates) at 99–100% coverage, so their *covariates* are unaffected; the risk-set *membership* is not, because a product enters the hazard sample only if a listed patent can be dated, and the products this drops are 88% foreign—so the demand-side sample, like

the citation sample, is domestic-route-skewed. The full sample-construction funnel and a characterization of the dropped set are in Appendix C; we return to the point when reading the foreign gradient (§7).

An external market-value measure: the National Reimbursement Drug List.

To test whether the challenge-targeting patterns of §7.4 reflect product market value rather than the covariates we model, we add one value proxy the register itself does not carry: inclusion in China’s 2024 National Reimbursement Drug List (国家医保目录, NRDL), the National Healthcare Security Administration’s (NHSA) formulary of drugs eligible for public reimbursement and the single largest determinant of a molecule’s addressable market in China. We extract the full list from the official NHSA notice (2024-11-28) and join it to the register’s products through the same molecule-name normalization used for the challenge stream, matching 66.3% of chemical products (63.2% of the challenge-attraction analysis sample). We record two indicators: whether the product is reimbursed (*in NRDL*) and whether it sits in the high-value *negotiated* tier—drugs whose prices NHSA sets through central negotiation, typically the newest and most IP-protected molecules. The measure is a current-status (2024-edition) cross-sectional flag—binary reimbursement status, not a continuous sales panel—so we use it to condition on value, not to measure it.

5 The Register at Birth: Supply Side

5.1 Quality selection into the register

Listing is selective, and the selection differs sharply by class. TCM products list a mean of 1.72 distinct patents, chemicals 2.09, and biologics 2.45; raw quality follows the same ordering (mean family forward citations 7.0, 22.5, and 79.4 respectively). The selection content is the headline: within ($\text{IPC4} \times \text{grant-year} \times \text{applicant-type}$) cells of the 1.56M-patent pharma denominator, listed patents carry a large citation premium over comparable

unlisted patents (Table 3), estimated by PPML (Santos Silva and Tenreyro, 2006)—the appropriate estimator for citation counts with many zeros (Cohn et al., 2022; Chen and Roth, 2024). Because the CNIPA merge is class-differential (§4), our *primary* specification is the coverage-matched estimate that restricts both listed and comparison patents to the domestic-filing route on which coverage is near-complete: it returns premia of +247% (chemical) and +990% (biologic), and because the excluded PCT national-phase patents are the register’s highest-cited members (§4), these are conservative lower bounds. Table 3 reports the larger full-sample premia alongside, for comparison; the cross-class gradient is the same under either, and is stable across the full robustness suite (Figure 4).

Table 3: Within-cell quality-selection premium (listed vs. comparable unlisted), PPML with (IPC4 $\times$ grant-year $\times$ applicant-type) fixed effects. Coefficients are log incidence-rate ratios; “Multiplier” is $\exp(\beta)$. $n = 226,736$; 199 cells; 20 IPC4 clusters. From the PPML re-estimation suite (RESULTS_DIGEST.md); 23-spec robustness suite in the online appendix.

	Family forward citations		Plain forward citations	
	β	Multiplier	β	Multiplier
Chemical	1.299***	3.67 $\times$	0.958***	2.61 $\times$
Biologic	2.420***	11.24 $\times$	1.267***	3.55 $\times$
TCM (comparison)	−0.029	0.97 $\times$	0.535***	1.71 $\times$

*** $p < 0.01$ (IPC4-clustered). TCM family premium: $p = 0.51$. The class

premia survive a wild-cluster bootstrap over the 20 IPC4 clusters

(Cameron et al., 2008): chemical $p = .006$, biologic $p = .05$, TCM $p = .65$;

and Romano–Wolf FWER control (Romano and Wolf, 2005): biologic $p = .0005$,

chemical $p = .019$, TCM $p = .55$.

The family-vs-plain divergence is the analytically interesting part: all three classes select on *domestic* technological impact (plain citations), but only chemicals and biologics additionally select on global-family breadth—the dimension on which Hatch–Waxman-style

market-defense calculus operates. The mechanism is structural: across distinct matched patents, the ratio of mean family to mean plain forward citations is 1.08 for TCM, 2.47 for chemicals, and 6.04 for biologics¹; TCM patents simply lack international family structure (Squicciarini et al., 2013; Sampat and Williams, 2019). The register thus selects globally-defended portfolios precisely in the classes the statutory machinery covers.

Subsidy-driven citation distortion and the cross-class gradient. A literature on Chinese patent statistics warns that subsidy and target policies have inflated filing volumes and citation counts, so that raw Chinese citation *levels* are a noisy quality proxy (Hu and Jefferson, 2009; Dang and Motohashi, 2015); Boeing and Mueller (2019) show specifically that domestic and self-citations carry an upward, subsidy-induced bias while only foreign citations track international quality benchmarks. This is a caution about levels, and our design is built to absorb it. We never interpret a level: identification is *within* (IPC4 $\times$ grant-year $\times$ applicant-type) cells, so any distortion common to listed and unlisted patents in the same technology–vintage–applicant cell differences out, and the object we interpret is the cross-class *gradient* (biologic > chemical > TCM), not any class’s absolute premium. A subsidy distortion could overturn that gradient only if it fell differentially on *listed relative to unlisted* patents within a cell, and did so in the biologic-over-chemical order—a pattern the subsidy literature does not describe. The family-citation measure adds a second layer of protection, drawing on the international search-report-style citations Boeing and Mueller (2019) find least distorted; and the biologic premium’s reading as strategic portfolio breadth rather than pure stay-defense is consistent with U.S. evidence that biologics patenting is driven by biosimilar-blocking thickets (Frakes and Wasserman, 2025).

¹These raw ratios partly reflect route composition: chemical and biologic listings include PCT/foreign-route filings, which carry the highest family-to-plain ratios, whereas TCM is 97.5% domestic-route. Restricting all classes to the TCM-comparable domestic-route subset, the ratios are TCM 1.08, chemical 1.71, biologic 4.18—the ordering is unchanged, and TCM’s near-unity ratio (essentially no international families) is invariant to route. Because unmatched chemical and biologic patents are disproportionately PCT national-phase (61–66% of the unmatched set), the lower CNIPA match rates for those classes (50.4% chemical, 36.8% biologic, against 97.5% TCM) drop their highest-ratio patents, so the matched-sample contrast is a *lower* bound—differential matching biases against, not toward, the cross-class gap.

The estimand is selection, not treatment: listing is a choice, and the within-cell premium measures *what originators choose to list*, not what listing does to a patent. One selection channel runs deeper than the listing choice: only patents on *approved* drugs can be listed, and approval correlates with the commercial success that drives citations, so the cell fixed effects bound but cannot fully purge an approval-selection component—we therefore read the premia as an *upper bound* on the listing-margin sorting. That is the object of interest for an anatomy—the register’s revealed sorting rule. Two bounds thus operate at once, and they are not in tension because they discipline different estimands: the class-differential coverage non-match strips the highest-cited foreign patents and biases the *measured* premium toward zero (the lower bound of §4), while approval-selection inflates it as an estimate of pure listing-margin sorting (the upper bound just stated)—and the cross-class gradient we interpret survives both. The premium survives randomization inference (Young, 2019) (2,000 within-cell relabelings), the coverage-matched domestic-route specification (§4), and a 23-specification robustness suite in the online appendix. Crucially, the cross-class *gradient*—the object we interpret—is not an artifact of differential family-citation coverage: in the coverage-matched specification the premia are +247% (chemical) and +990% (biologic), so the biologic-over-chemical ordering survives equal-footing matching.

Inference for the biologic premium. The biologic premium warrants a specific caution, because listed biologic patents are concentrated in a handful of technology classes: C07K alone holds 63% of them and two IPC4 classes hold 92%, so the effective number of treated clusters behind the biologic contrast is small. Leave-one-cluster-out estimates that drop each IPC4 class containing listed biologics leave the magnitude essentially unchanged (incidence-rate ratio 10.5–11.7; no single class drives it) but push the wild-cluster bootstrap p into the 0.05–0.12 range. We therefore treat the biologic premium’s magnitude as stable and its significance as borderline, and rest inference for the biologic arm on the randomization test (Young, 2019) rather than on cluster-count asymptotics.

5.2 Register composition and the evergreening margin

Self-certified listing systems invite structurally weak listings; the FTC’s 2023 campaign against improperly listed Orange Book patents is the regulatory expression of that concern ([Federal Trade Commission, 2023](#)), and the secondary-patent margin is where U.S. challenges concentrate ([Hemphill and Sampat, 2012](#)). China’s register allows a narrower set of listable categories than the Orange Book, which caps—but does not eliminate—the analogous margin.

The register’s composition is the first structural fact a linkage system reveals about itself. Among 948 listed products, 51.8% have live protection resting entirely on secondary patents—formulation, medical-use, or extract patents rather than the active compound. Class composition matters for interpreting that number: TCM products cannot list a compound patent by construction (the register’s primary categories—active-ingredient compound and sequence-structure—do not exist for TCM), so the economically meaningful figure is the chemical one: **45.2% of listed chemical products (209 of 462) carry no live compound patent**, and 43.1% never listed one at all. Among the 109 chemical products that list both patent types, the secondary patent outlives the compound in 81% of cases, extending nominal register protection by a **median 5.0 years** (maximum 19.2). A further 6.9% of listed chemical products retain no live listed patent of any kind.

Table 4: Live register protection by class, 27 May 2026 snapshot (distinct-patent level; CDE-stated expiry, 99.8% coverage). “Never primary” products listed no compound/sequence patent at any point; “orphaned” listed one that has since expired while a secondary patent remains live. TCM cannot list a primary category by construction.

	All (948)	Chem (462)	Bio (102)	TCM (384)
Never primary, live secondary	50.7%	43.1%	31.4%	65.1%
Orphaned (primary expired)	1.1%	2.2%	0%	0%
Secondary-only live protection	51.8%	45.2%	31.4%	65.1%
Has a live primary	30.5%	—	—	—
All listed patents expired	17.7%	6.9%	—	—

The forward-looking margin is starker than the snapshot. Among the 109 chemical products listing both patent types, the secondary patent outlives the compound in 81% of cases, with a median extension of 5.0 years (maximum 19.2)—assembled within the register’s first five years, where Orange Book evergreening accreted over decades (Feldman et al., 2022; Olsen et al., 2023; Alhiary et al., 2023). And the structure is already priced by challengers: chemical products with no compound patent are challenged at 41.7% (name band) against 19.5% for products with live compound protection—the cross-tabular form of the attraction result in §7.4. Unlike the FTC’s improper-listing cases, which turn on patents that do not claim the approved product—a scope question we cannot adjudicate—these are structural facts about what the register’s lawful categories protect; we therefore characterize composition and extension, and let the challenge gradient carry the economic content.

Stated versus actual legal status. The composition table reads protection off the register’s own stated expiry dates. A patent can die before that date—by invalidation, fee lapse, or abandonment—and a register that never de-lists will overstate protection. To measure this we attach INPADOC legal-event histories (via EPO Open Patent Services) to 1,986 of the 2,006 distinct register patents (99%; Appendix B). The register is 78% legally alive:

1,546 patents in force, 366 (18%) expired at natural term, **44 (2.2%) fully invalidated**, 21 lapsed for non-payment, and a handful ceased, abandoned, or never granted; a further 26 live patents carry partial invalidations. The stated-expiry field handles natural death perfectly: *zero* term-expired patents show a future stated expiry. What it never records is premature death. Of the 67 prematurely dead patents, **36 (54%) remain listed with stated expiries in the future**—nilotinib’s capsule patent (invalidated November 2024, stated expiry 2030), levothyroxine (invalidated 2021, stated 2033), copanlisib (fee-lapsed, stated 2034), befortinib (fee-lapsed 2025, stated 2035), dotinurad (invalidated 2025, stated 2038). Thirteen of the 36 sit in the CNIPA-matched majority, so this is a register-maintenance fact, not a coverage artifact. It is a governance failure, not evidence of entry deterrence: **84%** of the invalidated patents were challenged *before* they died (below), so sophisticated challengers evidently see through the stale display rather than being held off by it—the finding is about the register’s bookkeeping, not about demonstrated deterrence of generic entry. The bias this imparts to Table 4 is bounded and flagged rather than corrected: at most 36 of 2,006 patents (and their products) overstate live protection, and the correction requires product-level relinking we leave to the replication package.

How listed patents die. The invalidations are not background noise; they are the system’s enforcement margin at work—the administrative-invalidation analogue of the post-grant opposition channel that clears blocking patents and conditions follow-on entry (Gaessler et al., 2024; Galasso and Schankerman, 2015). All but two of the 44 full invalidations post-date the reform, peaking in 2024–25, and they follow the challenge stream: joining each dead patent’s registered drugs to the 17,599 challenge declarations, **98% of fully invalidated patents protected a drug that was challenged, 84% challenged before the invalidation** (median lag 21.0 months from first challenge; 19.2 months among chemical patents). The like-for-like contrast: 82% of invalidated *chemical* patents were challenged first, against a 29% ever-challenged baseline among active chemical patents. Partial invalidations show

the same pattern (74% challenged first). Fee lapses are the control group nature provides: only 14% of fee-lapsed patents’ drugs were ever challenged—originators quietly drop what nobody attacks, while challenged patents die at the review board. Two caveats keep this descriptive: invalidation petitioners are not observed (CNIPA proceedings can be initiated outside the linkage channel), and the join is at drug level. But the incidence gap and the time ordering support a reading that matters for the institution’s welfare accounting: *the linkage system’s teeth are administrative invalidation, not the courtroom stay*. The register’s failure is bookkeeping, not enforcement—invalidated patents stop binding but never stop being displayed.

6 Listing Dynamics

6.1 Accretion: returning to the register

Post-listing patent accretion—originators returning to add patents to a product’s listing—is the registration-side analogue of Hatch–Waxman evergreening ([Hemphill and Sampat, 2011, 2012](#)), and the dated event layer measures it as a duration. Naive “ever-updated” shares mislead because biologics entered the register roughly two years after chemicals and are right-censored early; Kaplan–Meier estimates put the three-year cumulative probability of a first update at **54% for biologics, 28% for chemicals, and 8% for TCM** (Figure 1). A Cox model ([Cox, 1972](#)) (chemical base) places the biologic update hazard *above* chemicals (HR = 2.05, $p < 0.001$) and TCM far below (HR = 0.22, 95% CI [0.15, 0.33], $p < 0.001$); cohort-stratified Cox and Weibull cross-checks preserve the ordering. An update here is a register 更新 event in a month *after* first listing; the listing-month rows, which record the initial multi-patent registration rather than a maintenance event, are excluded (retaining them roughly halves the estimated deficit and violates proportional hazards), with Efron handling of ties. Under this definition the proportional-hazards assumption is not rejected (Schoenfeld global $p = 0.64$), so the single hazard ratios summarize the update deficits fairly

over the whole follow-up. The cleavage is TCM versus the two statutory classes: chemicals and biologics accrete actively, TCM listings are set once and left.

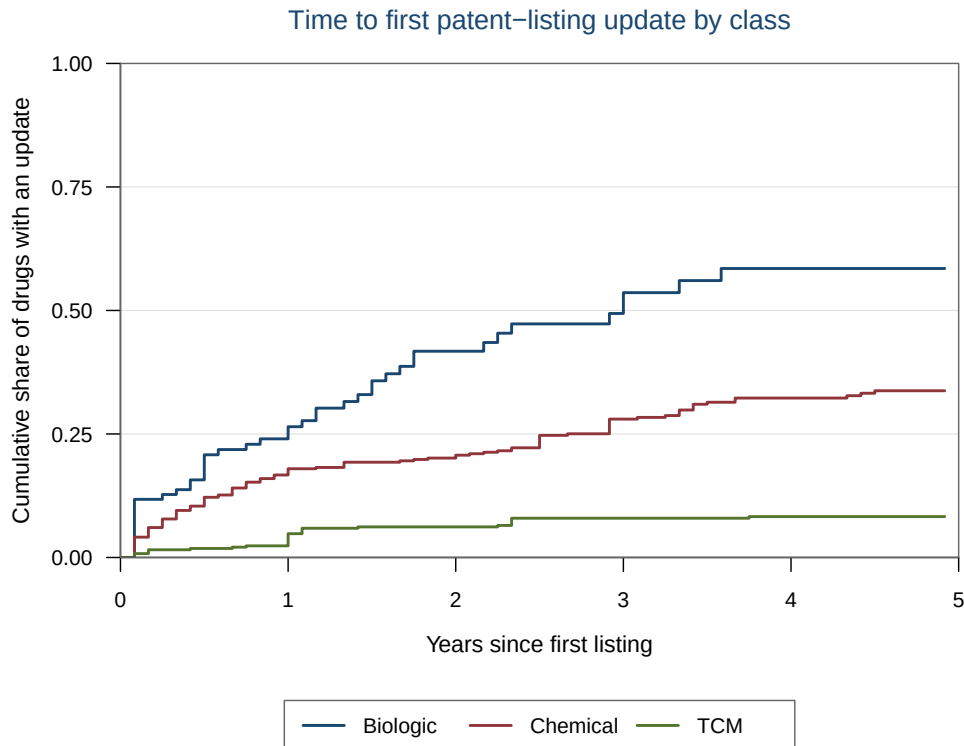


Figure 1: Kaplan–Meier cumulative incidence of a product’s first patent-listing update (更新) by class, right-censored 27 May 2026. An update is a 更新 event in a month after first listing; listing-month initial-registration rows are excluded and ties are handled by Efron’s method (proportional hazards not rejected, Schoenfeld global $p = 0.64$).

6.2 Updates are platform-driven, not challenge-responsive

A natural strategic hypothesis is defensive listing: originators add or update register entries *in response to* being challenged. The dated panel rejects this. In a monthly event study of register updates around a product’s first challenge, the pre-window is not flat (joint $p < 0.001$) and the post-challenge “spike” (+1.3pp over months 0–6) is indistinguishable from the spike produced when challenge dates are randomly reassigned across products challenged in the same quarter (randomization $p = 0.93$): updates cluster in calendar time alongside challenges because both ride the platform’s adoption ramp, not because products

respond to their own challenges. We report this as an anatomy fact: the register’s update dynamics are platform-driven (Figure 2); the timing facts themselves are in §7.1.

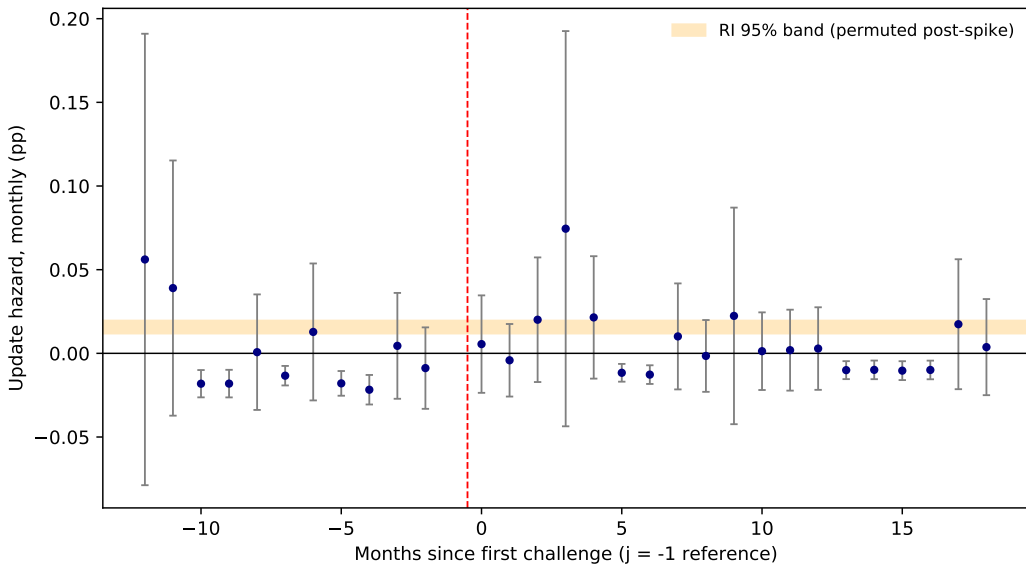


Figure 2: Monthly update hazard (percentage points) around a product’s first challenge ($j = -1$ reference; quarter and class FE; originator-clustered 95% CIs). The shaded band is the randomization-inference 95% interval of the permuted post-window spike: the observed post-challenge elevation sits *inside* what random reassignment of challenge dates within calendar quarters produces.

7 The Demand Side: Generic Challenges

7.1 Timing anatomy

The dated panel orders the two sides of the institution. Where a product is both listed and challenged, the listing comes first in 90.7% of cases, with a median gap of roughly 310 days—consistent with the register functioning as the statute intends: originators stake patent positions, and challengers respond about ten months later, plausibly the time to prepare an abbreviated application around a now-visible patent wall. The 9.3% of challenges that *precede* listing are themselves informative: 49 of the 137 ever-challenged chemical products were challenged by molecule name before their first register listing, so some originators list

after entry pressure has already materialized—late registration as an after-the-fact defensive response.

Challenge incidence is concentrated where the challenge apparatus operates (Table 5): under the strict approval-number join 9.7% of listed chemical products are challenged versus 2.0% of biologics and 1.8% of TCM; the name-band join puts the chemical rate at 29.7% while moving bio/TCM little. Chemical products account for 98.95% of all challenge declarations.

Table 5: Challenge incidence among listed products, by class and join rule. “Strict” joins challenges to products by normalized approval number; “band” joins by standardized product name (molecule level). The two rules bound true product-level incidence; all product-level results are reported under both (gate G1-C discipline).

	Listed products	Strict (%)	Name band (%)
Chemical	462	9.7	29.7
Biologic	102	2.0	2.9
TCM	384	1.8	1.6

7.2 Hazard of first challenge

Table 6 reports the discrete-time complementary-log-log hazard of first challenge for chemical products (product×month risk set; calendar half-year and listing-cohort fixed effects; originator- clustered SEs). Two pre-committed disciplines govern its reading. First, the duration-bin profile and the calendar ramp are *not separately identified* at five years of cohort variation—the profile’s shape depends on the fixed-effect scheme—so we report duration bins without interpretation. Second, the covariate associations are weak at the hazard level (the secondary-share coefficient, 0.43, has $p = .25$ analytically and $p = .17$ under randomization inference); the attraction evidence in this paper is cross-sectional (§7.4). What the hazard does establish robustly is the foreign-originator gradient—products of foreign originators face a hazard $e^{1.58} \approx 4.9$ times higher (band; strict: $e^{2.74} \approx 15$ times, both $p < 10^{-4}$; unchanged under ATC-1 therapeutic-class fixed effects, HR 5.6 and 21)—and the near-zero

bio/TCM baselines, which are the statutory point stated as levels.

Table 6: Discrete-time hazard of first challenge, chemical products (cloglog; calendar half-year + cohort FE; originator-clustered SEs). Band outcome: 88 events, 12,425 product-months; strict: 45 events, 13,440. Duration bins reported but not interpreted (not separable from the platform ramp). Time-varying covariates at the distinct-patent level.

	Name band		Strict join	
	β	(SE)	β	(SE)
$\ln(\text{listed patents})_t$	-0.397	(0.284)	0.214	(0.275)
Secondary share _t	0.434	(0.381)	0.330	(0.457)
Remaining life (yrs) _t	0.086	(0.106)	-0.212**	(0.104)
Remaining life _t ²	-0.004	(0.005)	0.010**	(0.005)
Foreign originator	1.580***	(0.353)	2.737***	(0.389)
Duration bins (6)	yes (uninterpreted)		yes	
Half-year + cohort FE	yes		yes	

** $p < 0.05$, *** $p < 0.01$. Sign disagreements across joins (patent count, remaining life)

are reported as measurement findings, not estimates (§7.4). The strict-join foreign

coefficient spans 2.74–2.81 across FE schemes (calendar-only, cohort-only, both);

the reported 2.737 (both) is the most conservative.

7.3 Enforceability news: the first linkage decision

Chemical challenge flows are roughly 70% higher in the quarters following the Beijing IP Court’s first linkage decision (Chugai v. Wenzhou Haihe, April 2022)—a Poisson level shift of +0.54 log points ($p = .007$) in a linear-trend interrupted time series, with a further +0.43 ($p = .075$) after the patent-term-extension regime took effect in 2024. We do not, however, read this as a discrete response to the decision, because the level shift fails two falsification checks. A placebo break placed one quarter *before* the decision (2022Q1) is itself significant

($+0.42$, $p = .02$), and sliding the break across all interior quarters ranks the true date only seventh of fifteen—chemical challenge filings rise smoothly across 2022–23 and recede in 2024–25, so many counterfactual break dates fit as well or better. Allowing the secular trend to curve (a quadratic ramp) reverses the shift to -0.48 ($p = .008$). The rise is thus a gradual post-reform ramp in challenge filing that we cannot separate from a discrete enforceability-news effect; biologic and TCM flows are too near zero to provide a control class. We therefore report the pattern as descriptive. The companion registration-flow plot in Figure 3 shows why no listing-flow causal interpretation follows either: biologic listings co-move with chemical listings around nominally chemical-only milestones.

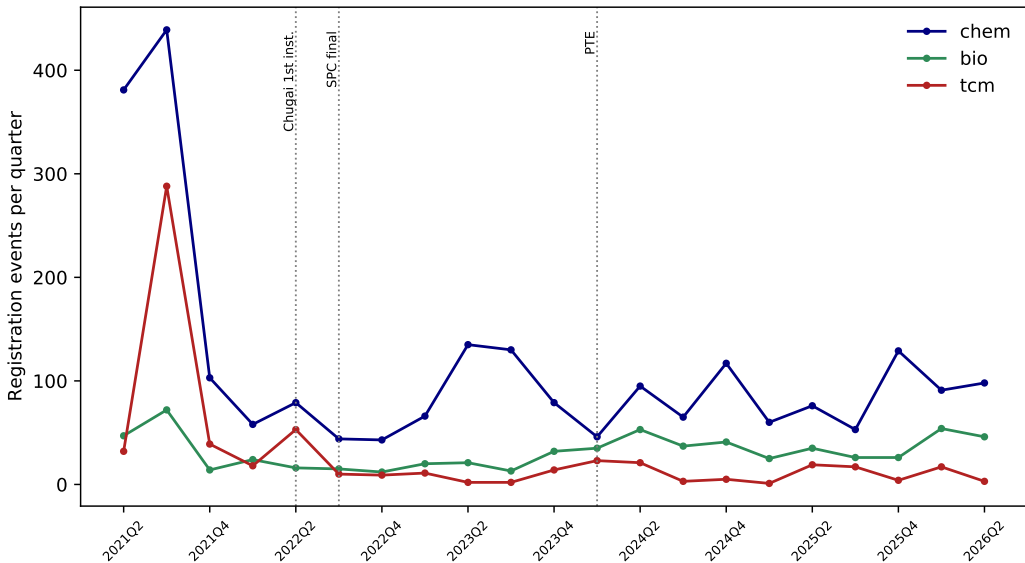


Figure 3: Register registration events per quarter by class, with judicial and regulatory milestones. The 2021Q3 spike is the register’s opening stock; the within-system falsification problem is visible directly: biologic flows co-move with chemical flows around “chemical-only” milestones.

7.4 What attracts a challenge

Table 8 asks which listed chemical products get challenged. Two facts are robust across joins, units, and fixed effects. First, *attraction*: a higher secondary-patent share raises the probability of challenge by 14–17pp (band, $p \leq .03$; molecule level +15pp, $p = .01$)—challengers

target products whose protection rests on formulation and use patents, the Hemphill–Sampat margin (Hemphill and Sampat, 2012), and the register-quality gradient of §5.2 (challenge rates of 41.7% on compound-free products vs. 19.5% otherwise) is the same fact in raw form.² Second, *foreign targeting*: products of foreign originators are 23–42pp more likely to be challenged, on both joins and at both units—the strongest single regularity on the demand side, and the cross-sectional form of the counterparty asymmetry (domestic challengers, foreign-heavy originators). Both survive Romano–Wolf FWER control across the demand-side family of tests (foreign and secondary share, each on both joins): foreign $p \leq .002$ on either join and secondary share $p = .002$ on the name band, so the multiplicity discipline applied to the selection premia (§5.1) is symmetric across the two sides of the register (Romano and Wolf, 2005; Clarke et al., 2020); both also clear randomization inference on the binary challenge outcome (exact $p < .001$), the appropriate check given the small event counts (Young, 2019). Nor is the foreign gradient a therapeutic-area artifact. Adding fixed effects for the product’s ATC-1 anatomical class—a therapeutic-area partition, so that foreign- and domestic-originated products are compared within the same broad class—leaves the foreign coefficient intact and slightly larger (band +0.284, $p < .001$; strict +0.415, $p < .001$; $n = 455$ of 462 products carry an assignable class), and the secondary-share attraction survives (+0.119, $p = .02$). A control for the number of register products sharing the molecule—a coarse within-register market-breadth proxy—moves the foreign coefficient by under one percent. The therapeutic-class result already removes the leading “foreign originators simply hold different drugs” confound;³ the one alternative it leaves open—that foreign products are simply more valuable and challengers chase value—we now test directly against an external reimbursement-value measure.

²The primary/secondary split is read from the platform’s own patent-type field; on the 8,351 rows typed on both sides it agrees with the IPC-implied class 88.9% of the time, leaving 11.1% disputed. Reassigning each disputed patent to its IPC-implied class, or dropping the 149 products those patents touch, leaves the secondary-share attraction coefficient in the 0.166–0.199 range (HC1 $p \leq .001$, randomization-inference $p \leq .003$) and the chemical secondary-only-protection share of Table 4 at 42.9–45.2%. The result does not turn on the disputed rows.

³ATC-1 classes were assigned algorithmically from active-ingredient names via international-nonproprietary-name stems (98.5% of products assignable); the crosswalk ships with the replication package.

Is foreign targeting a value proxy? A reimbursement-value decomposition. The sharpest remaining alternative to an origin premium is market value: foreign originators’ import-registered products may be more valuable, and challengers may chase value. We test this by adding the external NRDL controls of §4—an indicator for reimbursement (*in NRDL*) and for the high-value *negotiated* tier—to both the attraction LPM and the timing hazard (Table 7). The foreign coefficient does not shrink; it *rises*. In the attraction LPM it moves from +0.252 to +0.306 (band) and +0.405 to +0.430 (strict); in the hazard it moves from +1.580 to +1.867 (band) and +2.737 to +2.937 (strict), every foreign coefficient significant at $p < 0.01$ under both HC1 and 2,000-draw randomization inference. The value controls are informative, not inert: *in NRDL* is positive throughout (reimbursed, larger-market products are challenged more), while the newest, most IP-protected *negotiated* tier is challenged *less* on the band margin. The reason the foreign coefficient rises rather than falls is suppression: foreign originators are *under*-represented on the NRDL (46% reimbursed against 65% for domestic originators), yet reimbursement itself predicts challenge, so omitting NRDL status biased the foreign coefficient *toward* zero. Value-targeting and foreign-targeting are thus separate, partially offsetting margins: challengers do chase reimbursed value, but that channel works *against* the foreign signal rather than generating it. The foreign gradient is an origin premium in challenge risk conditional on value, not a market-value artifact—though NRDL is a binary reimbursement flag, and a continuous sales or procurement-volume panel would remain the sharper value control.

Table 7: Foreign-vs-value decomposition: the foreign coefficient does not shrink when an external NRDL reimbursement-value control is added—it strengthens. Columns (1)/(3) repeat the base attraction and hazard specifications; columns (2)/(4) add the *in NRDL* and *negotiated-tier* indicators of §4.

	Band outcome		Strict outcome	
	(1) Base	(2) +NRDL	(3) Base	(4) +NRDL
<i>Panel A. Attraction (cross-sectional LPM, Pr(challenged), listed chemical products)</i>				
Foreign originator	0.252***	0.306***	0.405***	0.430***
	(0.074)	(0.071)	(0.070)	(0.066)
In NRDL		0.405***		0.128***
		(0.048)		(0.028)
NRDL negotiated tier		−0.398***		−0.008
		(0.054)		(0.039)
Observations	462			
<i>Panel B. Timing (discrete-time cloglog hazard of first challenge, calendar×cohort FE)</i>				
Foreign originator	1.580***	1.867***	2.737***	2.937***
	(0.353)	(0.347)	(0.389)	(0.412)
In NRDL		1.073***		1.346**
		(0.355)		(0.653)
NRDL negotiated tier		−1.302***		0.012
		(0.325)		(0.360)
Product-months	12,425 (88 events)		13,440 (45 events)	

** $p < 0.05$, *** $p < 0.01$. NRDL = 2024 National Reimbursement Drug List

(official NHTSA source; molecule-name join, 66% match on chemical products).

Panel A: OLS LPM, HC1 SEs; Panel B: cloglog hazard, calendar-half-year + cohort

FE, originator-clustered SEs. Suppressed controls: log listed patents, secondary

share, remaining life and its square. RI (2,000 perm., Panel A): foreign $p < 0.01$.

By contrast, the patent-*count* coefficient flips sign across challenge measures (-0.13 band vs. $+0.04$ strict at the product level; -0.09^* vs. $+0.08^{**}$ at the molecule level), and the flip is *not* a mechanical name-join artifact: it is invariant to controls for the number of products sharing a molecule name and for name length (band $-0.135 \rightarrow -0.131$, $p < 0.01$ throughout), and the band match rate *falls* with patent count rather than rising as a spurious-match story would predict. It reflects instead *differential coverage*: the strict approval-number join fires cleanly for import-registered (foreign) products but misses many domestic-registration entries, and the products it misses are disproportionately low-patent—the 100 “band-only” products average 1.38 listed patents and are 7% foreign, against 2.08 and 57% for products matched on both keys. Because the strict join is thus a foreign-skewed, higher-patent subsample, neither sign identifies deterrence—the Ellison–Ellison question (Ellison and Ellison, 2011) is *indeterminate in this register*, and we make no deterrence claim in either direction. Remaining life shows no robust option-value hump. Even with the external NRDL reimbursement-value control now in hand, these remain conditional correlations rather than causal effects, and are worded as such.

Table 8: Probability of challenge, listed chemical products (LPM, HC1 SEs). Product level: $n = 462$ (137 challenged on the name band, 45 strict); molecule level: $n = 332$ (products aggregated by standardized name; covariates and outcome at the same unit). Probit signs match (not shown).

	Product level (band)		Molecule level (band)
	no FE	IPC4 FE	n -products control
ln(listed patents)	-0.135^{***} (0.036)	-0.130^{***} (0.036)	-0.089^* (0.046)
Secondary share	0.166^{***} (0.049)	0.138^{**} (0.063)	0.146^{**} (0.058)
Remaining life	0.010 (0.011)	0.009 (0.011)	-0.003 (0.016)
Foreign originator	0.252^{***} (0.074)	0.233^{***} (0.078)	0.182^* (0.096)
Strict-join sign	patent count + / rem. life $-^{**}$ / sec. share + / foreign +		

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Full strict columns and probit in appendix.

8 What the Register Does Not Show

A register observed from birth shows selection and targeting clearly; its *dynamics* are still dominated by institutional adoption, and we close by assembling what the data refuse to support, because each refusal was established by a pre-committed test rather than by absence of effort. The gate protocol fixing these tests—the G1-series referenced in the table captions—is a dated document in the replication package, so which results were pre-specified is verifiable rather than asserted. *No defensive listing*: the event-study pre-window around first challenge is not flat (joint $p < 0.001$) and the post-window “response” is mechanical (§6)—updates follow the platform, not the product’s own challenge. *Deterrence indeterminate*: the patent-count gradient flips sign between the strict and name-band challenge measures, and the flip survives every name-multiplicity control (so it is *not* a join artifact); it traces instead to the strict join’s differential, patent-correlated coverage of domestic versus import products (§7.4). With the strict join a non-random subsample in patent count, neither sign is interpretable as deterrence, and we decline any deterrence claim from this register. *Statutory machinery not separable*: biologic listing flows respond to the first chemical-drug linkage decision at least as strongly as chemical flows (+1.12 vs. +0.79 log points; robust to ± 3 -quarter local windows), so the chem-only stay cannot be assigned causal credit for any flow response. *Enforceability news not datable*: the post-reform rise in chemical challenge volume fails its placebo-in-time test—the first linkage-decision quarter ranks only seventh of fifteen candidate break dates, a pre-decision break is equally significant, and the level shift reverses under a flexible trend—so the challenge ramp cannot be pinned to the decision either. *Duration dependence not separable*: with five years of staggered cohorts, the months-since-listing profile is not identified apart from the platform ramp; we report hazard levels and decline to interpret shapes. Each null sharpens what the next five years of data—and the quarterly re-scrapes already scheduled—can resolve.

9 Conclusion

Observed from birth, China’s linkage register already reproduces the two robust regularities of forty years of Orange Book evidence—quality- selective listing and challenges that chase weak secondary protection on visible products—while adding two facts the U.S. literature cannot see: the protective backbone is thin from day one (45% of chemical products listed without a live compound patent; five years of median secondary extension), and the system’s challenge volume responds to enforceability news within quarters. For policy, the combination of self-certified listing and a thin compound backbone is the structural precondition for the improper-listing problem the FTC now polices in the U.S. ([Federal Trade Commission, 2023](#))—though whether China’s listings are lawful-but-weak or improper in scope is a question this register cannot adjudicate. What we can show is that China’s narrower listable categories cap that margin but do not close it, that the challenge gradient we document shows generics are already pricing register weakness, and that the integrity problem is no longer hypothetical: the register already displays three dozen invalidated or lapsed patents with stated expiries reaching 2038, even as the invalidations themselves—84% preceded by challenges on the same drugs—show the system’s administrative enforcement margin working. For research, the nulls define the agenda: whether thick walls deter, whether defensive listing emerges, and what the statutory machinery causes are questions that the register’s adolescence—and the scheduled re-scrape vintages—will progressively identify. The welfare quantity these facts bear on but cannot yet price—the effective market life a thin, secondary-heavy backbone confers, and the time-to-generic-entry it buys ([Hemphill and Sampat, 2012](#); [Panattoni, 2011](#))—awaits the downstream entry and price outcomes a later vintage will supply. What its infancy already shows is that the *static* grammar of Hatch–Waxman—selective listing and quality-sorted challenge, though not the dynamic responses—travels: within five years, under a different legal system and a far weaker reward, originators list what is globally defended and challengers attack what is locally weak.

A Robustness of the selection premium

Estimator and transform. The cross-class gradient is invariant to the count model and transform: negative binomial returns chemical +244%, biologic +986%, TCM -4% ($p = 0.50$); $\log(m+y)$ OLS is flat across $m \in \{0.5, 1, 2\}$, and inverse-hyperbolic-sine gives the same picture (chemical 0.86, biologic 1.69, TCM 0.02).

Design-based matching. Coarsened exact matching (Iacus et al., 2012) augmenting each cell with a backward-citation quartile returns chemical +0.69 ($p = 0.012$), biologic +1.32 ($p < 0.001$), and an economically negligible TCM coefficient (+0.05). Entropy balancing (Hainmueller, 2012) confirms the chemical premium at an attenuated 0.50 and the TCM near-null (0.09); for biologics, listing selection is so extreme that moment-balancing is infeasible (maximum standardized imbalance $1.53 \rightarrow 1.54$)—itself a diagnostic.

Quantile structure. Two-step fixed-effects quantile estimates (Canay, 2011) rise from the 25th to the 90th percentile for chemicals ($0.33 \rightarrow 1.32$) and biologics ($0.86 \rightarrow 2.96$) while flat-to-negative for TCM ($0.04 \rightarrow -0.13$): a distributional shift, not a right-tail artifact.

Coverage bounds and age-at-listing. The PCT omission biases chemical/biologic premia toward zero (§4); the coverage-matched estimate is a conservative lower bound. TCM’s older listed patents (median 8.9 years from grant vs. 4.4 chemical) do not generate the family null: the design conditions on grant year within cells, and the grant-to-listing lag does not differ by class once grant year is absorbed; older patents have had longer to accumulate citations, which works *against* a TCM null. The full 23-specification grid, randomization-inference distributions, and denominator-sensitivity table (36–112 listed per 100,000 across pharma-IPC definitions) are in the replication package’s results digest.

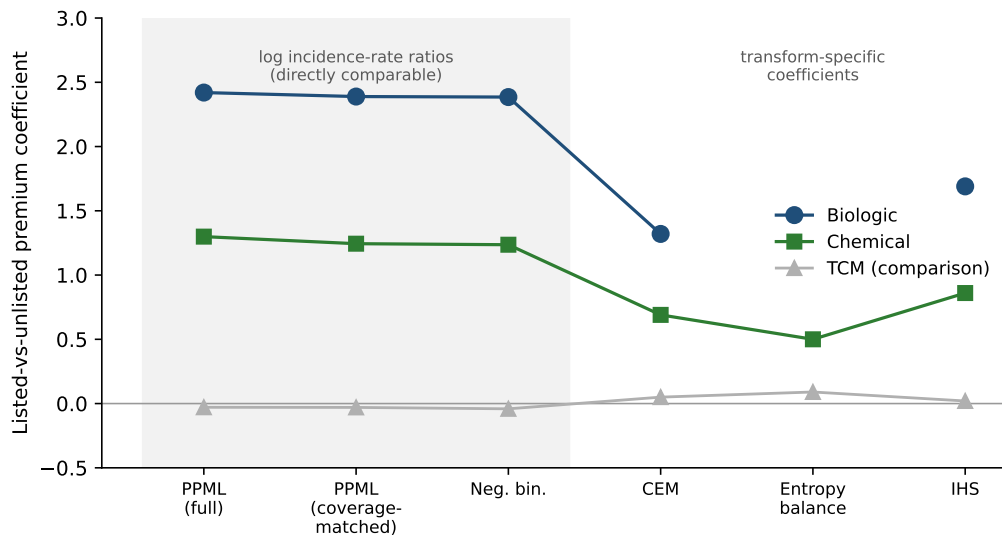


Figure 4: The cross-class selection gradient is invariant to how coverage and functional form are handled. Points are the listed-vs-unlisted premium coefficient by class across six specifications. The first three (full-sample PPML, the coverage-matched domestic-route PPML, and negative binomial) are log incidence-rate ratios and are directly comparable; the last three (CEM, entropy balancing, IHS-OLS) are transform-specific and comparable in sign and ordering only. Across all six, the biologic premium exceeds the chemical premium and both are clearly positive, while TCM sits at zero—so the gradient we interpret is not an artifact of the coverage correction or the count model. Biologic entropy balancing is omitted (moment-balancing infeasible; §A).

B Measuring the unmatched register patents

Sources and coverage. The 779 register patents the CNIPA merge leaves without usable bibliographic records (the PCT national-phase misses of §4 plus six matched records lacking classification fields) are measured directly from two independent sources. *Google Patents* supplies family-level forward citations and assignee data for 760 (98%); *EPO Open Patent Services* (OPS v3.2) supplies DOCDB biblio, INPADOC family membership, and INPADOC legal-event histories—for the unmatched patents and for the full register (1,986 of 2,006 resolved, 99%), with legal events matched to the listed publication itself rather than pooled across family siblings. On the patents both sources resolve, legal-status classifications agree 92.1%; disagreements are dominated by term expirations that Google Patents still displays as

active, so the paper uses INPADOC status throughout and Google Patents only for citation counts.

Citation calibration. Google Patents and CNIPA count citations on different bases. A pilot of 47 matched patents measured both ways gives Spearman $\rho = 0.90$ and a stable GP-to-CNIPA ratio of 0.55, which we use to place the supplement on the CNIPA scale:

Table 9: Family forward citations: register patents inside vs. outside the CNIPA merge. Matched column on CNIPA counts; unmatched on Google Patents counts, raw and calibrated (GP/0.55). Jurisdictions from INPADOC families.

	Matched (CNIPA)		Unmatched			Median
	<i>n</i>	mean	<i>n</i>	mean (raw GP)	mean (calibrated)	jurisdictions
Chemical	495	25.6	552	61.1	110.4	27
Biologic	113	89.3	196	171.8	310.3	27
TCM	449	6.8	12	3.8	6.9	1

The excluded chemical and biologic patents are thus 4.3 and 3.5 times more family-cited than their matched counterparts—the direct measurement behind the lower-bound statement in §4—and they are the register’s global families: 84% (chemical) and 94% (biologic) of unmatched patents have INPADOC families spanning ten or more jurisdictions, against a register-wide chemical median of 17. The twelve unmatched TCM patents are domestic and lightly cited, consistent with the class’s filing structure (§5.1). Raw OPS responses, the event-code classifier, and the invalidation–challenge join are in the replication package.

ρ -sensitivity. The lower-bound *direction*—that excluded patents are more family-cited than matched ones—does not depend on the calibration constant ρ . Recomputing the unmatched-versus-matched ratio across $\rho \in [0.42, 1.0]$ leaves the excluded chemical and biologic patents more family-cited than the matched ones at *every* value on the grid, including $\rho = 1.0$, i.e. raw Google Patents counts with no calibration at all (chemical 2.4 $\times$, biologic

1.9 $\times$). The direction could reverse only if Google Patents *overcounted* the CNIPA scale by more than 1.9 $\times$ (i.e. $\rho > 1.9$), which contradicts the measured undercount ($\rho \approx 0.55 < 1$). The specific ρ therefore governs the *magnitude* of the excess, never its sign. The single class that is knife-edge is TCM (ratio ≈ 1.0 at $\rho = 0.55$, reversing below $\rho \approx 0.56$)—but TCM is excluded from the lower-bound argument, its twelve unmatched patents being domestic and lightly cited, so this boundary does not bear on the chemical and biologic bounds we report.

Transportability of the pilot. One limit of the calibration is transparent, and we state it rather than bury it. The 47-patent dual-measured pilot is drawn disproportionately from small, domestic families—median INPADOC family size 2, median one jurisdiction—whereas the unmatched patents to which ρ is applied are large multinational families (median family size 74 for unmatched chemical patents and 82 for biologic, spanning a median of roughly 30 jurisdictions). Because the Google-Patents-versus-CNIPA citation divergence plausibly scales with family breadth, the *calibrated* point magnitudes (the 110.4 and 310.3 columns of Table 9) are less certain than the raw-count *direction*; a family-size-stratified pilot is the natural fix at revision.

C Sample construction and attrition

Table 10 traces both pipelines from the raw scrape to the estimation samples, by class. The patent funnel runs from listing rows to the CNIPA-matched subset that carries citation outcomes; the product funnel runs from the register’s products to the chemical hazard sample and its event counts. Every node reproduces the counts stated in §4 and §7.

Table 10: Sample-construction funnels (CONSORT), by class. Panel A: the patent pipeline from raw listing rows to the CNIPA-matched subset used for citation outcomes. Panel B: the product pipeline from register products to the chemical hazard sample. Dashes denote nodes defined only for chemical products.

	Total	Chemical	Biologic	TCM
<i>Panel A. Patent pipeline</i>				
Listing rows	7,905	5,572	1,354	979
Distinct patents	2,006	1,127	318	561
CNIPA-matched (has record)	1,233	569	117	547
<i>Panel B. Product pipeline</i>				
Register products	1,821	1,100	313	408
Dated risk set	948	462	102	384
Chemical hazard sample	462	462	—	—
ever challenged (band)	137	137	—	—
ever challenged (strict)	45	45	—	—
hazard events (band)	88	88	—	—
challenged pre-listing	49	49	—	—

Who the risk-set restriction drops. The 873 register products excluded from the 948-product risk set for carrying no *datable* listed patent are **88.0% foreign-originated**, against 9.4% in the retained risk set, and are chemical/biologic-heavy (73%/24%/3% by class). The drop mechanism is precisely the PCT national-phase non-match that removes foreign-route patents from the CNIPA layer (§4): these products carry a mean 1.77 listed patents in the raw register but zero that can be dated. The hazard sample—like the citation sample—is therefore a domestic-route-skewed subsample, so the foreign-originator gradient of §7 is, if anything, estimated against a conservative composition rather than inflated by one.⁴

⁴The 88-after / 49-before split of the 137 ever-challenged chemical products uses listing-*month* versus challenge-month granularity; at *day* granularity the split is 85 after / 52 before, because three products are

The pre-listing-challenged products. The 49 chemical products challenged *before* they complete listing are the thinnest and most secondary-heavy in the register—a mean of 1.29 listed patents and a 75% secondary share, against 2.07 patents and a 56% secondary share for never-challenged chemical products. The left-truncated set is thus exactly the weakest protection, which corroborates rather than threatens Fact 3 (challenges chase weak, secondary protection; §7.4): the products attacked earliest are those with the least to defend.

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